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MATH 246A

Homework 4
Due: TBD

Problem 7.1

Suppose $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ is a power series with nonzero coefficients. Show that its radius of convergence R is given by

$$R = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right|$$

if the limit exists.

Solution. Let $L = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right| \in [0, \infty]$, and set $w = z - z_0$. Write $t_n = a_n w^n$. We show the series $\sum t_n$ converges for $|w| < L$ and diverges for $|w| > L$, which means $R = L$.

If $w = 0$ the series trivially converges, so assume $w \neq 0$. Since every $a_n \neq 0$, every $t_n \neq 0$, and

$$\left|\frac{t_{n+1}}{t_n}\right| = |w| \left|\frac{a_{n+1}}{a_n}\right| = \frac{|w|}{\left|\frac{a_n}{a_{n+1}}\right|}.$$

Taking $n \to \infty$ (with the conventions $\frac{|w|}{0} = \infty$ and $\frac{|w|}{\infty} = 0$),

$$\lim_{n\to\infty} \left|\frac{t_{n+1}}{t_n}\right| = \frac{|w|}{L}.$$

Case $|w| < L$. Then $\frac{|w|}{L} < 1$ (this includes $L = \infty$, where the limit is 0). By the Ratio Test, $\sum t_n$ converges absolutely.

Case $|w| > L$. Then $\frac{|w|}{L} > 1$ (this includes $L = 0$, where the limit is ∞). So there is N with $|t_{n+1}| > |t_n|$ for all $n \geq N$, hence $|t_n| \geq |t_N| > 0$ for $n \geq N$. Thus $t_n \not\to 0$, and the series diverges (equivalently, by the Ratio Test).

Hence the series converges for $|z - z_0| < L$ and diverges for $|z - z_0| > L$, so $R = L$.

Problem 7.2

Find the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n z^n$, where
(a) $a_n = (2i)^n/(n+i)^2$,
(b) $a_n = n^2 + 5in + 1$,
(c) $a_n = n^{\log n}$.

Solution.

Problem 7.3

Show that the series $\sum_{n=1}^{\infty} \frac{z^n}{n}$ converges for $z \in \mathbb{C}$ if and only if $|z| \leq 1$ and $z \neq 1$.

Solution. Checking the radius of convergence:

$$\begin{aligned} R &= \lim_{n\to\infty}\left|\frac{1/n}{1/(n+1)}\right| \\ &= \lim_{n\to\infty}\left|\frac{n+1}{n}\right| \\ &= 1 \end{aligned}$$

Thus for all $z \in \mathbb{D}$, the series converges. Now we need to examine the behavior of the series when $z \in \partial\mathbb{D}$. For $z = 1$, the series is the harmonic series which diverges. Now we need to understand how the series behaves $z = e^{i\theta}$ for $\theta \in (0, 2\pi)$.

Consider

$$\begin{aligned} |S_n| &= \left|\sum_{k=1}^{n} e^{in\theta}\right| \\ &= \left|\frac{e^{i\theta}\left(1-e^{in\theta}\right)}{1-e^{i\theta}}\right| \\ &= \left|\frac{1-e^{in\theta}}{1-e^{i\theta}}\right| \\ &\le \frac{\sqrt{2}}{|1-e^{i\theta}|} \end{aligned}$$

Thus S_n is uniformly bounded.

$$\begin{aligned} \sum_{k=1}^{n}\frac{e^{ik\theta}}{k} &= \sum_{k=1}^{n}\frac{1}{k}(S_k - S_{k-1}) \\ &= \sum_{k=1}^{n}\frac{1}{k}S_k - \sum_{k=1}^{n}\frac{1}{k}S_{k-1} \\ &= \sum_{k=1}^{n}\frac{S_k}{k} - \sum_{k=0}^{n-1}\frac{S_k}{k+1} \\ &= \frac{S_n}{n} - \frac{S_0}{m} + \sum_{k=1}^{n-1} S_k\left(\frac{1}{k} - \frac{1}{k+1}\right) \\ &= \frac{S_n}{n} + \sum_{k=1}^{n-1} S_k\left(\frac{k+1-k}{k(k+1)}\right) \\ &= \frac{S_n}{n} + \sum_{k=1}^{n-1} S_k\left(\frac{1}{k(k+1)}\right) \end{aligned}$$

The partial sums converge (This was just the dirichlet test).

Problem 7.4

Suppose that the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ and $g(z) = \sum_{n=0}^{\infty} b_n z^n$ have radii of convergence $R_1 > 0$ and $R_2 > 0$, respectively. For $n \in \mathbb{N}_0$ define

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Show that the power series $h(z) = \sum_{n=0}^{\infty} c_n z^n$ (the so-called *Cauchy product* of f and g) converges for all $|z| < R := \min\{R_1, R_2\}$ and that $h(z) = f(z)g(z)$ for all such z.

Solution. Fix a arbitrary $w \in B(0, R)$. Since $|w| < R_1$ and $|w| < R_2$, we know that

$$A = \sum_{k=1}^{\infty} |a_n w^n| \text{ and } B = \sum_{k=1}^{\infty} |b_n w^n| \text{ converge.}$$

Let

$$f_n(z) = \sum_{k=0}^{n} a_n z^k$$
$$g_n(z) = \sum_{k=0}^{n} b_n z^k$$
$$h_n(z) = \sum_{k=0}^{n} c_k z^k.$$

We know that $f_n \to f$ and $g_n \to g$ converge locally uniformly on $B(0, R)$. We want to show that $f_n g_n - h_n \to 0$.

$$\begin{aligned}(f_n g_n - h_n)(w) &= \sum_{k=0}^{n} \sum_{l=0}^{n} a_k b_l w^{k+l} - \sum_{k=0}^{n} \sum_{l=0}^{k} a_l b_{k-l} w^k \\ &= \sum_{\substack{k,l \le n \\ k+l > n}} a_k b_l w^{k+l}\end{aligned}$$

Thus

$$\begin{aligned}|f_n(x)g_n(x) - h_n(x)| &\le \sum_{\substack{k,l \le n \\ k+l > n}} \left|a_k b_l w^{k+l}\right| \\ &\le \sum_{k=\lceil \frac{n}{2} \rceil}^{\infty} |a_k| \left(\sum_{l=1}^{\infty} |b_l||w^l| \right) |w^k| + \sum_{l=\lceil \frac{n}{2} \rceil}^{\infty} |b_l| \left(\sum_{k=1}^{\infty} |a_k||w|^k \right) |w|^l \\ &= A \sum_{k=\lceil \frac{n}{2} \rceil}^{\infty} |a_k||w|^k + B \sum_{l=\lceil \frac{n}{2} \rceil}^{\infty} |b_l||w|^k\end{aligned}$$

We can bound the tails with $\frac{\varepsilon}{2A}$ and $\frac{\varepsilon}{2B}$ and we are done. Thus $\lim_{n\to\infty}(f_n(w)g_n(w) - h_n(w)) = 0$ which means that

$$h(w) = \lim_{n\to\infty} h_n(w) = \lim_{n\to\infty} f_n(w)g_n(w)$$

Problem 7.5

Let $R > 0$, and suppose that the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ converges for all $z \in B := B(0, R)$.
(a) Show that the power series converges uniformly on each compact subset of B.
(b) For $n \in \mathbb{N}_0$ and $z \in B(0, R)$ define

$$S_n(z) := \left| \sum_{k=0}^{n} a_k z^k \right|^2.$$

Show that on each compact subset of B the function sequence $\{S_n\}$ converges uniformly to $|f|^2$.
(c) Show that for each $0 \le r < R$ *Parseval's formula* is valid:

$$\frac{1}{2\pi} \int_0^{2\pi} \left| f(re^{it}) \right|^2 \mathrm{d}t = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}.$$

Solution.

1. Let K be a compact subset of $B(0, R)$. The $x \to |x|$ map is continuous and must achieve its maximum on K, ie $\exists k \in K$ such that $\forall x \in K$, $|k| > |x|$. But note that $k \in K \subseteq B(0, R)$, thus $r = |K| < R$. Let $S \in (r, R)$.

   $$\frac{1}{S} \ge \frac{1}{R} = \limsup_{n \to \infty} |a_n|^{\frac{1}{n}}$$

   this means for all but finitely many $n \in N$ we have that

   $$\frac{1}{S} \ge |a_n|^{\frac{1}{n}} \Rightarrow S^{-n} \ge a_n.$$

   Note that for $k \in K$, $|k| < r$. Thus we have that for $x \in K$:

   $$\sum_{k=0}^{\infty} |a_n||x|^k \le \sum_{k=0}^{\infty} S^{-k} r^k.$$

   By the M test

   $$\sum_{k=0}^{\infty} \left| a_n x^k \right| \text{ converges uniformly on } K \Rightarrow \sum_{k=0}^{\infty} a_n x^k \text{ converges absolutely uniformly on } K$$

2. Let K be a arbitrary compact subset of B.

$$\begin{aligned}
\left|S_n - |f|^2\right| &= \left|\left|\sum_{k=0}^{n} a_k z^k\right|^2 - |f|^2\right| \\
&= \left|\left(\left|\sum_{k=0}^{n} a_k z^k\right| - |f|\right)\left(\left|\sum_{k=0}^{n} a_k z^k\right| + |f|\right)\right| \\
&\leq \left(\left|\sum_{k=0}^{n} a_k z^k\right| + |f|\right)\left|\sum_{k=0}^{n} a_k z^k - f\right| \\
&\leq \left(\sum_{k=0}^{n}|a_k z^k| + \sum_{k=0}^{\infty}|a_k z^k|\right)\left|\sum_{k=0}^{n} a_k z^k - f(z)\right| \\
&\leq 2\sum_{k=0}^{\infty}|a_k z^k|\left|\sum_{k=0}^{n} a_k z^k - f(z)\right| \\
&\leq 2M\left|\sum_{k=0}^{n} a_k z^k - f(z)\right| \\
&\leq \varepsilon
\end{aligned}$$

Problem 7.6

(a) Show that the power series

$$\sum_{n=0}^{\infty} \frac{i^{n+1}}{(n+1)!} z^n$$

converges for all $z \in \mathbb{C}$.
(b) Let g be the function represented by the power series in (a). Find a simple expression for $g(z)$ if $z \neq 0$.
(c) For $r > 0$ let $\alpha_r : [0, 2] \to \mathbb{C}$ be the path defined by

$$\alpha_r(t) = \begin{cases} re^{i\pi t} & t \in [0, 1] \\ (2t - 3)rt \in [1, 2]. \end{cases}$$

Show that $\int_{\alpha_r} g(z)\,\mathrm{d}z = 0$. *Hint:* This follows from a general theorem that will be proved later, but show this directly.
(d) Show that $\lim_{r\to\infty} \int_{[-r,r]} g(z)\,\mathrm{d}z = i\pi$.
(e) Show that $\int_0^{\infty} \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$.

Solution.

1.

$$\begin{aligned}
R &= \lim_{n\to\infty}\left|\left(\frac{i^{n+1}}{(n+1)!}\right)\left(\frac{(n+2)!}{i^{n+2}}\right)\right| \\
&= \lim_{n\to\infty}|n+2| = \infty
\end{aligned}$$

2.
$$\begin{aligned}\sum_{n=0}^{\infty} \frac{i^{n+1}}{(n+1)!} z^n &= i \sum_{n=0}^{\infty} \frac{(iz)^n}{(n+1)!} \\ &= i \sum_{n=1}^{\infty} \frac{(iz)^{n-1}}{n!} \\ &= i \left(\sum_{n=1}^{\infty} \frac{1}{iz} \frac{(iz)^n}{n!} \right) \quad (z \neq 0) \\ &= \frac{1}{z}(e^{iz} - 1)\end{aligned}$$

3. Let
$$\begin{aligned}G(z) &= \sum_{n=0}^{\infty} \frac{i^{n+1}}{(n+1)(n+1)!} z^{n+1} \\ &= \sum_{n=1}^{\infty} \frac{(iz)^n}{n(n!)}.\end{aligned}$$
$G(z)$ has a radius of convergence of ∞, furthermore $G' = g$. Thus since the path is a loop, by FTC we know
$$\int_{a_r} g \, \mathrm{d}z = 0$$

4.

Problem 7.7

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$.
(a) Show that for each $t \in [0, 2\pi]$ the function P_t defined as
$$P_t(z) = \frac{e^{it} + z}{e^{it} - z} \quad \text{for } z \in \mathbb{D}$$
can be represented by a power series that is centered at $z_0 = 0$ and converges on $\mathbb{D}$.
(b) Let $u : \partial\mathbb{D} \to \mathbb{R}$ be a continuous function. Show that the function $f : \mathbb{D} \to \mathbb{C}$ defined as
$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{e^{it} + z}{e^{it} - z} u(e^{it}) \, \mathrm{d}t \quad \text{for } z \in \mathbb{D}$$
is holomorphic. *Hint:* Find a power series representation of f on $\mathbb{D}$.

Solution.

Problem 7.8

Define $\gamma : [0, 2\pi] \to \mathbb{C}$ as $\gamma(t) = e^{it}$ for $t \in [0, 2\pi]$.
(a) Show that

$$\frac{1}{2\pi i}\int_\gamma \left(w-\frac{1}{w}\right)^n \frac{\mathrm{d}w}{w} = \begin{cases} 0 & n \in \mathbb{N}_0 \text{ odd} \\ (-1)^{n/2}\binom{n}{n/2} & n \in \mathbb{N}_0 \text{ even.} \end{cases}$$

The *Bessel function* $J_0 : \mathbb{C} \to \mathbb{C}$ is defined as

$$J_0(z) = \frac{1}{2\pi i}\int_\gamma \exp\left(\frac{z}{2}\left(w-\frac{1}{w}\right)\right)\frac{\mathrm{d}w}{w}$$

for $z \in \mathbb{C}$.

(b) Show that J_0 is represented by the power series

$$J_0(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{z}{2}\right)^{2n}$$

convergent for all $z \in \mathbb{C}$ and is hence an entire function, that is, holomorphic on $\mathbb{C}$.

Solution.

Problem 7.9

(a) Show that the function J_0 defined in Problem 7.8 satisfies *Bessel's differential equation*

$$zJ_{0''}(z) + J_{0'}(z) + zJ_0(z) = 0$$

for $z \in \mathbb{C}$.

(b) Suppose $f(z) = \sum_{n=0}^{\infty} a_n z^n$ is a power series that converges for $z \in \mathbb{C}$ and the function f represented by this power series satisfies Bessel's differential equation. Show that if $f(0) = 1$ and $f'(0) = 0$, then $f = J_0$. *Hint:* Find a recursive relation that the coefficients a_n must satisfy.

Solution.

Problem 7.10

The periodic oscillations of a thin membrane (such as a drum) can be modeled as follows: in rest position the membrane occupies a region U in the x-y plane. Its movement under periodic (undampened) oscillations is described by a function $u = u(x, y, t)$ measuring the (small) deflection of the membrane in the z-direction in a three-dimensional x-y-z coordinate system over the point $(x, y) \in U$ at time t. The motion of the membrane is then governed by the two-dimensional *wave equation*

$$u_{tt} = u_{xx} + u_{yy}$$

with the boundary condition $u = 0$ on ∂U (with a suitable unit of time depending on material constants of the membrane). Here we denote by a subscript differentiation with respect to the variables t, x, y.

(a) Suppose u is a solution of the wave equation of the form $u(x, y, t) = f(t)v(x, y)$ with sufficiently smooth functions f and v, where $f \neq 0$ is a periodic function with $f(0) = 0$. Show that then there exist constants $A \in \mathbb{R}$ and $\lambda > 0$ such that

$$u(x, y, t) = A\sin(\lambda t)v(x, y),$$

where

$$\Delta v + \lambda^2 v = 0.$$

(b) Suppose that the membrane is circular, say $U = \mathbb{D}$. Suppose that $u(x, y, t) = f(t)g(r)$ is a solution of the wave equation, where f is as in (a) and g only depends on the radius r in polar coordinates in the x-y-plane. Show that g then satisfies the differential equation

$$rg''(r) + g'(r) + \lambda^2 rg(r) = 0.$$

(c) Find solutions g of the equation in (b) in terms of λ and the Bessel function J_0 (see Problem 7.9). Which condition on λ guarantees that $u(x, y, t) = f(t)g(r)$ as in (b) for one of these solutions g satisfies the boundary condition $u = 0$ on $\partial\mathbb{D}$?

> *Solution.*